



Ishaan Vitthala



Class of 2028

I'm a driven high school student passionate about medicine, biology, and original research that solves real-world problems. I've explored areas like cancer biology, epigenetics, and biotechnology, and I'm always eager to keep learning and pushing boundaries.



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Marquette High School
Chesterfield MO

EXTRACURRICULARS

NSDA Speech and Debate

Achieved 850+ NSDA points = Top 0.5%

Special Distinction Award

MHSHSAA State qualifier for Duet Acting -1%

MHSHSAA State Qualifier for Lincoln Douglas, Placed 11th at state-1%

FIRST Tech Challenge Team Action Robotix

Received Sustain Award at league tournament

First Place Inspire award and Winning alliance at superqualifiers

Won sustain award and robot playoffs at FTC state missouri

DECA

Awarded 1st place in Startup Business District Award

Achieved 2nd Place in Business Management and Administration District Award

Delegate for Results Political Group

Actively participated in initiatives to help pass the Read Act and other educational reforms, advocating for improved literacy and educational standards.

Financial Manager for Sugarcane Juice bar-chesterfield mo

Oversaw daily operations and full financial management of a family-owned juice retail business, scaling it with parental investment support to achieve \$10,000+ in monthly profit.

Scioly

District 3rd place team, qualified to state. 7th place at state.

HOSA

Made Nationals for Creative Problem solving HOSA, 3rd place

Placed Top 5 in researched Persuasive Speaking and Writing for HOSA

EDUCATION

UW GPA: 4.0

W GPA: 4.6

Relevant Coursework

AP Physics 1

Honors Chemistry

Honors Biology

Biomedical Science

Honors Algebra 2

SKILLS

LANGUAGES:

- ENGLISH: NATIVE OR BILINGUAL PROFICIENCY
- SPANISH: PROFESSIONAL WORKING PROFICIENCY
- TELEGU PROFESSIONAL WORKING PROFICIENCY

TECHNICAL SKILLS:

- PROFICIENT IN RESEARCH METHODOLOGIES,
- DATA ANALYSIS, AND PROJECT MANAGEMENT.
- EXPERIENCED IN PUBLIC SPEAKING, LEADERSHIP, AND TEAMWORK.
- SUFFICIENT IN AUTOCAD
- SUFFICIENT IN PYTHON AND MACHINE LEARNING/ AI
- PARTLY SUFFICIENT IN HTML AND R CODING

CPR/FIRST AID CERTIFIED
STOP THE BLEED CERTIFIED

PAST RESEARCH PROJECTS

CELL: Cancer Epigenome Language Learner – A Novel Transformer-Based Framework for Multi-Modal Epigenomic Cancer Classification and Prognosis Prediction

- I developed CELL (Cancer Epigenome Language Learner), a transformer-based deep learning model that integrates DNA methylation, histone modifications, and gene expression data to classify cancer types and predict patient outcomes. Trained on data from over 11,000 patients across 33 cancers, CELL achieved 94.2% classification accuracy and significantly improved survival prediction. The model also uncovered 336 novel epigenetic biomarkers and provided interpretable insights into key genomic regions. This work demonstrates the potential of AI to advance personalized cancer diagnostics and treatment.

Digital Resurrection: Predicting the Biochemical Function of Extinct Human Enzymes Using Ancient DNA and AI

- I led a project combining ancient DNA analysis with AI to predict the biochemical functions of extinct human enzymes from Neanderthals and Denisovans. Using a hybrid ancestral reconstruction and deep learning approach, we achieved 94% accuracy and uncovered enzymes with enhanced thermostability and metabolic flexibility. The study successfully resurrected 47 ancient enzymes and revealed evolutionary shifts in human metabolism. This work offers a new framework for molecular de-extinction and has exciting applications in drug discovery and biotechnology.

Teacher and Student Ambassador for Smart Kids Ghana

Taught math and biology to **students in Ghana**, helping strengthen their academic foundation and future opportunities. Served as a student ambassador, leading fundraising efforts to provide computers and improve digital access for the school. Combined education and advocacy to empower underserved students with both knowledge and technology.

[Ishaan Blog link](#)

PROGRAMS

Saint Louis University Doisy College of Health Sciences (20% acceptance rate)

Explored a range of medical fields, including radiology and hematology, through direct engagement with SLU faculty, including **8 PhD professors**. Earned the **Best Scholar Award**—granted to **only 4 out of 22 students**—for high engagement and initiative throughout the program.

Harvard Undergraduate Microbiology Society (6% acceptance rate)

Selected to join a prestigious student-run initiative aiming to spread microbiology awareness through outreach. Contributed to published articles, blogs, and podcast content to make microbiology accessible to the broader community.

Published an article about [Bacteriophage Symphony: Engineering Phages as Live Biocomputers to Reprogram Microbial Communities on medium news](#).

New York Academy of Sciences

Selected as part of a highly competitive cohort (**≈10% acceptance rate**) for the New York Academy of Sciences Junior Academy, a global program connecting top students with STEM professionals and researchers. Collaborated internationally to tackle real-world scientific challenges through research-driven problem solving, critical analysis, and interdisciplinary teamwork.

BEst Healthcare institute WashU

Selected BHI scholar (month-long program; \$500 stipend). Shadowed radiologists at BJC HealthCare, gaining exposure to diagnostic imaging, clinical workflows, and patient-centered decision-making within an academic medical system.

CURRENT PROJECTS

A Network-Aware Computational Framework for Cancer Drug Repurposing Using Graph Embeddings

I am currently working on a computational cancer drug-repurposing project that models cancer as a broken gene interaction network rather than a single-gene disease, using network-level analysis to identify existing drugs that can restore healthy gene behavior. By building and validating a reproducible R pipeline on large public cancer expression and pharmacogenomic datasets, I am testing whether network-guided approaches outperform traditional gene-by-gene methods in predicting effective cancer treatments.

PFAS Sentinel: A 3D-Printed Handheld Breakthrough-Risk Monitor with Regenerable Capture Cartridges for Point-of-Use Water Protection.

PFAS are “forever chemicals” that contaminate drinking water and are extremely hard to remove, and most filters today give users no real way to know when they stop working. Our project solves this by building a handheld, 3D-printed cartridge system that not only captures contaminants but also monitors filter performance in real time. Water is pushed through a packed media chamber (Purigen or activated carbon), where contaminants are trapped as the water flows through the tiny spaces between particles. We integrated an optical sensing system (LED + photoresistor + Arduino) that measures subtle changes in light through the cartridge window, which correlates to how “loaded” the media is and helps detect breakthrough risk. To validate this, we run controlled experiments comparing fresh, exhausted, and regenerated cartridges while recording sensor data and water samples. The project goes further by implementing a closed-loop regeneration step, where the media is chemically restored and retested to quantify how much performance returns. We use replicated trials, control groups, and statistical comparisons (like breakthrough curves and percent capacity recovery) to ensure the results are scientifically rigorous. Overall, this turns a simple filter into a smart, testable, and partially reusable water treatment system, which is why it stands out as a first-place-level innovation. **I earned the \$529 MOST scholarship from the STL Academy of Sciences for this project. Presented it at the Illinois High School Research Symposium, and got accepted to present at IYRC this summer.**

Universal Topological Phase Transition Signatures Preceding Human Physiological Instability Across Multimodal Stress Systems

Physiological instability is often detected only after symptoms or performance decline appear. This project investigates whether orthostatic, cognitive, and physical stress share a common pre-instability mathematical signature: loss of physiological complexity. Using multimodal wearable time-series data, I extract nonlinear features such as detrended fluctuation exponents, Hurst exponent, entropy measures, fractal dimension, and topological summaries from persistent homology. These features are tested through baseline modeling, change-point analysis, nonparametric statistics, cross-condition comparison, and supervised classification. The goal is to determine whether stable systems transition through a shared critical state before failure, enabling earlier wearable-based detection of instability and intervention in real time. **I presented this project at IMSSRS**

VOLUNTEER WORK

Student Council (STUCO)

- Contributed 30+ hours organizing and setting up various activities around the school and community, fostering school spirit and engagement.

Key Club

- Volunteered 20+ hours, participating in community service projects to support local initiatives.

Red Cross Club

- 50+ hours under the club and **Treasurer position**, helping manage funds and getting outreach programs

FBLA (Future Business Leaders of America)

- Received the **CSA 50 Hour Volunteer Award** for dedicated community service.

FIRST LEGO League Volunteer Mentor – Action Robotix

Volunteered 100+ hours mentoring a team of 5th graders in robotics, strategy, and problem-solving for the FLL Robot Game. Guided the team to a 2nd place finish in the Robot Game at the district competition through weekly coaching sessions, technical support, and collaborative teamwork.